

Micro-Filter Air Purifier

Innovation dossier

1. What is it?

A concept for the real 'Filter cost / replacement (weak real signal)' friction, built with the DISTRIBUTE design move - one large system -> several smaller systems.

2. Physical architecture

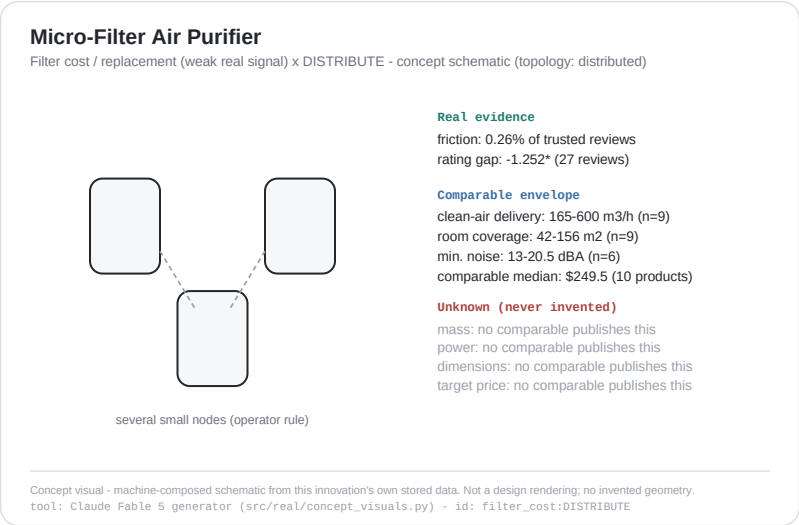


Figure: [Concept visual](#)

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.
artifact visual-filter_cost:DISTRIBUTE - /concept-visuals/filter_cost_DISTRIBUTE.svg

Product archetype

- room scale:** Unknown - no real data in this pipeline for this field.
 - mobility:** Multi-unit - operator DISTRIBUTE turns one system into several smaller ones.
 - size class:** Unknown - no real data in this pipeline for this field.
 - power class:** Unknown - no real data in this pipeline for this field.
 - noise ambition:** Unknown - no real data in this pipeline for this field.
 - maintenance burden:** Unknown - no real data in this pipeline for this field.
 - filter service dependency:** High - this concept directly targets filter cost/frequency friction.
 - connected:** Unknown - no real data in this pipeline for this field.
 - form factor:** distributed
- Design rule - each populated field follows from the operator's own fixed definition, never from evidence

System architecture

Unknown - no system-architecture dataset exists; the archetype above is a design rule from the operator's own definition.

Method choice

DISTRIBUTE - One large system -> several smaller systems.
Method choice - authored design vocabulary

3. Why it exists

Reality Observed

0.26% of trusted reviews (27 reviews across 15 real products) carry this friction, with a -1.252* average rating gap. Signal: Filter cost/frequency is a real complaint with no matching peer-reviewed literature.

Households whose real reviews carry this friction: 27 reviews across 15 real products (70.4% verified purchase, 2012-04-24-2023-07-21).

Unknown - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

Transformation Method choice

METHOD CHOICE, not evidence: the authored operator table assigns DISTRIBUTE - "One large system -> several smaller systems." - to this friction. Analyst design judgment, applied deterministically.

Product consequence Feasibility precedent

Feasibility precedent (from the trend corpus, not a consumer-evidence consequence): No real technical or regulatory document in this corpus addresses subscription logistics or bundled-pricing design - like customer_service, this is a real business-model change, not a technology gap, and this pipeline has no real Versuni pricing/operations capability data to assess it. Rating reflects LOW technical complexity only; commercial feasibility is genuinely unassessed here.

4. Evidence and lineage

Parent paths

No parent paths - this innovation cites no tension or assumption paths; its grounding is the friction evidence in section 3, why it exists.

evidence ids: taxonomy:filter_cost

category assumptions: No category assumption shares a paper with this theme's signal evidence.

Design DNA

F consumer friction (taxonomy:filter_cost) Present

Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator Method choice

Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (filter_cost) Present

Filter cost/frequency is a real complaint with no matching peer-reviewed literature.

T scientific tension Missing / unverified

No real research tension shares a paper with this theme's signal evidence.

R rival gap Missing / unverified

No real competitor-weakness data clears the white-space threshold for this theme.

C versuni capability Missing / unverified

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption Missing / unverified

No category assumption shares a paper with this theme's signal evidence.

E economic condition Present

Real price-weighted exposure computed for this theme (\$7,390.66).

5. Comparable products

comparable set: 10 verified official Versuni/Philips air-purifier families (room-scale)
clean-air delivery rate (CADR): 165-600 m3/h across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)
room coverage: 42-156 m2 across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)
minimum noise: 13-20.5 dBA across 6 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

6. Engineering envelope

metric	value from verified comparables	epistemic type
clean-air delivery rate (CADR)	165-600 m3/h (n=9 comparables)	Observed comparable
room coverage	42-156 m2 (n=9 comparables)	Observed comparable
minimum noise	13-20.5 dBA (n=6 comparables)	Observed comparable
target mass	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target power	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	Unknown

7. Economics

price-weighted exposure: \$7390.66

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$249.5 (n=10 products) - not this concept's price

MEDIAN real listed price across the 10 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

target price: Not modelled - No concept target-price model exists - only the comparable market median (\$249.5, 10 real products) as reference. A target price would be invented.

8. Prototype

State: Concept visual

Concept visual means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

9. Experiment and next learning

Next experiment

Real internal margin/logistics data for a filter-subscription model would let this split into a proper technical-feasibility rating plus a separate, currently-missing commercial-feasibility rating.

Kill criterion

No authored kill criterion yet for this innovation - only the formal case (see formal_case.abandon_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

10. What could kill it?

Where this idea stands: rejected - graveyard

Killed in the Magic funnel: Not enough real complaint evidence - 0.26% is below the 0.5% floor.

kill criterion: No authored kill criterion yet for this innovation - only the formal case (see formal_case.abandon_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

Uncertainties

- No real Versuni pricing, margin, or subscription-operations data - commercial feasibility is genuinely unassessed, not just unfavourably assessed.

Critic dimensions

Evidence	Reject	Consumer Pain evidence-sufficiency gate failed - no real, materially-prevalent friction to build against.
Economic	Challenge	Real price-weighted exposure (\$7,391) is below the corpus median (\$16,113) - a real but comparatively modest signal.
Competitive	Needs evidence	No real rival-weakness data recorded for this friction theme.
Human	Needs evidence	No real usability/human-factors evidence exists in this pipeline for this concept.
Physical	Needs evidence	No real lab/engineering feasibility data exists in this pipeline for this concept.
Versuni fit	Needs evidence	No real organizational-capability assessment exists in this pipeline for this concept.
Timing	Needs evidence	No real market-timing/window evidence exists in this pipeline for this concept.
Robustness	Needs evidence	No real sensitivity/robustness study exists in this pipeline for this concept.
critic overall: Reject		

Criteria results

Needs evidence: 35 - Pass: 9 - Challenge: 3 - N/A: 3 - Kill: 2

E3	Challenge	Signal state SINGLE_SOURCE_FAMILY - real, but not yet independently corroborated.
H2	Kill	gate_passed=False
V8	Challenge	Operator DISTRIBUTE may add surface area to manage - unverified.
C1	Kill	Not white space.
F3	Challenge	medium

Lineage and provenance

generated by: src/real/innovations_real.py
magic run input sha256: 0b3f9862e0e270a295eacd709182802e981090e817f4f1072000e37aa83f150c
funnel snapshot: d5d03b0d3856721f04d7d1c4ee7f2fe2a7d1df64ac4d2d4130bede07d2b13f61
figure source: /concept-visuals/filter_cost_DISTRIBUTE.svg (generated 2026-08-31T18:00:05.652995+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)
data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)
dossier generator: src/real/generate_innovation_dossiers.py